

MIT653 – Assessment 2-1

Assessment Title: Literature Review Report

Volume of Assessment: 2,500-word Report

Total: 100 points

Weighting: 20%

Artificial Intelligence Rules of Engagement: Level 3 – AI Tools are allowed for editing and improving students' work. In any case, if you use generative AI, you must precisely acknowledge how it was used and keep a copy (e.g. screenshots) of all the interactions that you had with the AI tools, as you may be asked to provide this evidence, after the submission of your task.

Submission Guidelines:

- **Submission Due: 16 August 2026, Saturday 23:59pm**
- Submit your Literature Review Report via Moodle as a written document in a standard format (e.g., PDF or Word document).

Objective:

The objective of this assignment is to conduct a comprehensive literature review on a given research topic in the field of Information Technology. This task will help you gain insights into the current state of the given research topic, identify emerging issues, and critically analyse recent research in the field.

Instructions:

1. **Research Topic**
 - The research topic is the topic that you are assigned in Week 1 class.
2. **Literature Search:**
 - Conduct an extensive literature search using academic databases, journals, conference proceedings, and reputable online sources. Aim to find recent research articles (published within the last 5-10 years) that are relevant to your chosen topic.
3. **Literature Review:**
 - Write a literature review that summarises and synthesises the key findings and insights from the selected articles. Your literature review should:
 - Introduce the topic and its importance in the given research topic.
 - Review and summarise the main arguments, methodologies, and results of the selected articles.
 - Identify trends, patterns, and emerging issues in the given research topic.
 - Highlight gaps or areas where further research is needed.
4. **Critical Analysis:**
 - Conduct critical analysis of the literature, including strengths and weaknesses of the studies reviewed.

- Discuss the implications of the research findings for the field of the given research topic.
- Consider ethical considerations and limitations in the studies you have reviewed.

5. Organization and Presentation:

- Organise your literature review logically, with clear headings and subheadings.
- Ensure proper citations and references following **IEEE citation style**.
- **Required Structure of the Report:**
 - Abstract
 - Table of Content
 - Introduction & Background
 - Purpose & Aims of the Literature Review
 - Literature Review Process
 - Main Body of the Report
 - Conclusions
 - References

Assessment Criteria:

Your literature review will be assessed based on the following criteria:

1. **Relevance to the Chosen Topic** (10 points)
 - The extent to which your literature review is relevant to the given research topic.
2. **Literature Search** (15 points)
 - The comprehensiveness and effectiveness of your literature search in identifying recent and relevant research articles.
3. **Quality of Literature Review** (25 points)
 - The quality of your literature review, including the clarity and coherence in presenting the review; the depth of evaluation, analysis, synthesis, and organisation of information from the selected literature; the logical flow and structure of ideas.
4. **Critical Analysis** (25 points)
 - Your ability to critically analyse the literature, including identifying strengths, weaknesses, trends, and gaps in the research.
5. **Proper Citations and References** (15 points)
 - Proper use of citations and references, following the chosen citation style consistently.
 - A reasonable number of references should be included (**20-30 references**)
6. **Overall Presentation** (10 points)
 - The overall quality of your writing, including adherence to the required structure, as well as grammar, spelling, and formatting.

MIT653 – Assessment 2-1

Literature Review Report

The Effectiveness of Generative AI in Creating Personalised Video and Audio Content

Abstract

Generative Artificial Intelligence (GenAI) has rapidly changed digital content creation, particularly in video, music, and speech generation. Recent advances in text-to-video, text-to-music, and speech synthesis allow users to create personalised digital content with less technical knowledge and reduced production time. However, effectiveness cannot be evaluated only through realistic outputs. Content quality, personalisation, production efficiency, authenticity, copyright, privacy, misinformation, bias, and human creativity must also be considered. This literature review examines recent research across three main dimensions: content quality and personalisation, production efficiency, and ethical challenges. The literature indicates that GenAI can improve accessibility, scalability, and production speed, but limitations remain in consistency, controllability, semantic accuracy, authenticity, and ethical governance. The review therefore argues that effective use of GenAI is best understood as human-AI collaboration, combining automated generation with human creative direction, quality control, and ethical judgement.

1. Introduction and Background

Generative Artificial Intelligence has become an important technology for digital content creation. Unlike traditional artificial intelligence systems that mainly analyse or classify existing information, generative AI can produce new text, images, music, speech, and video. Recent advances in deep learning and generative models have made it increasingly possible for users to create digital media from relatively simple instructions. This development has significant implications for creative industries because content production has traditionally required specialised technical skills, expensive software, and substantial time.

Generative AI is particularly relevant to video and audio production. Text-to-video systems can transform textual descriptions into video sequences, while text-to-music and speech-generation systems can create audio based on user instructions. Models such as Make-A-Video, Imagen Video, VideoPoet, MusicLM, MusicGen, and Voicebox demonstrate rapid development in these capabilities. These technologies provide opportunities for advertising, education, entertainment, social media, and personalised communication.

The development of these technologies also changes the role of human creators. Instead of producing every component manually, creators can increasingly use AI to generate initial content, modify existing material, and automate repetitive tasks. This can potentially reduce production costs and increase productivity. However, generated video may contain temporal inconsistencies or unrealistic physical behaviour, while generated audio may raise concerns regarding voice identity, authenticity, and unauthorised imitation.

The ethical dimension is equally important. Barnett's systematic review of generative-audio research found that negative impacts were discussed much less frequently than positive applications, despite concerns including fraud, deepfakes, and copyright infringement. This suggests that technological development can progress faster than systematic consideration of its social consequences.

Therefore, this review examines the effectiveness of generative AI in creating personalised video and audio content from the perspectives of quality and personalisation, production efficiency, and ethical responsibility.

2. Purpose and Aims of the Literature Review

The purpose of this literature review is to critically examine research on generative AI for personalised video and audio content creation. The review addresses three research questions:

RQ1: How does generative AI affect the quality, realism, creativity, and relevance of personalised video and audio content?

RQ2: How does generative AI improve the efficiency, accessibility, and scalability of digital content production?

RQ3: What ethical challenges are associated with AI-generated video and audio, particularly regarding authenticity, copyright, privacy, misinformation, bias, and human creativity?

The review aims to synthesise findings across studies rather than simply summarising individual papers. It also identifies common trends, strengths, limitations, and research gaps.

3. Literature Review Process

The literature search focused primarily on research published from 2020 to 2026, consistent with the assessment requirement to prioritise recent literature. Academic and reputable online sources were considered. Search terms included “generative AI”, “AI-generated video”, “text-to-video generation”, “generative audio”, “AI music generation”, “speech generation”, “voice cloning”, “personalised digital content”, “content quality”, “production efficiency”, “deepfake”, “AI ethics”, and “copyright”.

The literature was organised into four themes: technological development, content quality and personalisation, production efficiency, and ethical challenges. Priority was given to peer-reviewed journal articles and established conference proceedings. Technical papers describing major generative models were included where they directly demonstrated the capabilities or limitations of current systems.

A literature matrix was used to compare the selected studies by focus, methodology, key findings, limitations, and relevance to the research questions. Ten key studies were selected for the matrix because they provide balanced coverage of video, audio, efficiency, quality, and ethics. Additional reputable online and academic sources can be used in the final reference list to support regulatory, industry, and background claims.

3.1 Literature Review Matrix

Table 1 summarises ten key studies used as the foundation for the thematic synthesis.

No .	Study	Focus	Method	Key Findings	Limitations / Gap	Relevance
1	Singer et al. (2022)	Make-A-Video / video	Model development and evaluation	Demonstrates text-to-video generation and improved video	Quality and temporal consistency remain challenging.	Video quality

				synthesis capability.		
2	Ho et al. (2022)	Imagen Video / video	Diffusion-model development and evaluation	Demonstrates high-definition video generation with strong controllability.	Computational demands and generation limitations.	Video quality
3	Kondratyuk et al. (2024)	VideoPoet / multimodal media	Multimodal model development and evaluation	Supports generation across text, image, video, and audio modalities.	Multimodal evaluation and control remain challenging.	Personalisation
4	Agostinelli et al. (2023)	MusicLM / music	Generative model and evaluation	Generates music from textual descriptions and supports conditioning.	Fine-grained controllability remains limited.	Personalised audio
5	Copet et al. (2023)	MusicGen / music	Model development and human/automatic evaluation	Provides controllable text-to-music generation and improved generation quality.	Human refinement can still be required.	Quality and control
6	Le et al. (2023)	Voicebox / speech	Experimental model evaluation	Demonstrates scalable and efficient multilingual speech generation.	Voice misuse and safety require attention.	Production efficiency
7	Barnett (2023)	Generative audio ethics	Systematic literature review	Reviews 884 generative-audio papers and identifies	Ethical research is less developed than	Ethics

				under-discussed negative impacts.	technical research.	
8	Mirsky & Lee (2021)	Deepfakes	Survey	Shows the growing challenge of creating and detecting synthetic media.	Generation and detection create an ongoing arms race.	Authenticity
9	Hagendorff (2024)	Generative AI ethics	Comprehensive scoping review	Maps a broad range of ethical issues across generative AI.	Practical governance mechanisms need further development.	Ethics
10	Amankwah-Amoah et al. (2024)	Creative industries	Conceptual/research agenda	Highlights productivity and innovation benefits alongside disruption to creative work.	More empirical evidence is needed on long-term labour effects.	Efficiency and human labour

Table 1. Literature review matrix of ten key studies.

4. Main Body of the Report

The matrix reveals a clear progression from technical capability toward questions of practical effectiveness and responsible use. The literature can be synthesised around content quality and personalisation, production efficiency, ethical challenges, and research gaps.

4.1 Generative AI for Video and Audio Content Creation

Recent research demonstrates rapid progress in generative video and audio. Make-A-Video and Imagen Video show that textual descriptions can be converted into visual sequences, while VideoPoet extends the approach to multimodal generation. In audio, MusicLM and MusicGen demonstrate text-conditioned music generation, while Voicebox demonstrates scalable speech generation.

The studies collectively show that GenAI is moving from single-purpose generation toward more flexible multimodal content creation. However, technical capability does not automatically mean that generated content is suitable for professional use. Effectiveness depends on quality, consistency, controllability, relevance, and the amount of human intervention required.

4.2 Content Quality and Personalisation

Content quality is a central measure of generative AI effectiveness. Video models have improved visual fidelity and semantic alignment, while audio models have improved naturalness and controllability. Nevertheless, quality is multidimensional. A video may appear realistic in individual frames but contain temporal or physical inconsistencies across a sequence. Similarly, generated audio may be technically clear but fail to reflect the intended emotion, style, or identity.

Personalisation introduces an additional requirement: the output must match the user's intended purpose. A personalised educational video, for example, needs to be relevant to the learner, while personalised marketing content needs to reflect audience preferences. Therefore, relevance and controllability should be evaluated alongside technical realism.

The matrix also shows a gap between automated metrics and human judgement. Future evaluation should combine technical measures with user assessment of usefulness, relevance, realism, and trustworthiness.

4.3 Production Efficiency and Accessibility

Production efficiency is one of the strongest potential advantages of GenAI. Traditional video and audio production can involve planning, recording, editing, sound design, visual effects, and post-production. Generative systems can automate parts of these workflows and reduce the technical expertise required for initial content creation.

Voicebox, MusicLM, MusicGen, and multimodal video systems illustrate how AI can make content generation more scalable. Users can generate multiple versions of content for different audiences without producing every version manually. This is particularly relevant to advertising, education, social media, and multilingual communication.

However, generation speed should not be treated as equivalent to total production efficiency. Human review, editing, fact checking, copyright checks, and ethical approval may still be necessary. The true efficiency gain should therefore consider the complete workflow from initial prompt to final approved content.

4.4 Ethical Challenges

The ethical literature identifies several risks associated with synthetic video and audio.

First, authenticity and deepfakes are major concerns. Increasingly realistic synthetic media can make it difficult for audiences to distinguish genuine events from generated content. Deepfakes can support misinformation, impersonation, fraud, and reputational harm. Detection systems are also challenged by continuous improvements in generation technology.

Second, copyright and ownership remain unresolved issues. Generative systems may be trained on large collections of existing media, creating questions about training data, authorship, ownership, and the rights of original creators. These issues are especially relevant to music and voice generation.

Third, voice cloning creates privacy and consent concerns because a person's voice can function as an identifiable aspect of their identity. Responsible systems therefore require consent, transparency, and mechanisms that allow individuals to control the use of their voice.

Fourth, generative systems can reproduce biases from training data. Personalisation can unintentionally reinforce stereotypes or produce unequal representations. This is particularly important in education, advertising, and public communication.

Finally, GenAI may change creative labour. Automation can reduce repetitive production tasks but may also alter employment and professional roles. The literature therefore suggests that human-AI collaboration may be more sustainable than complete replacement of human creators.

4.5 Critical Analysis and Research Gaps

The literature identifies several important gaps. First, technical development is more mature than research into negative consequences. Barnett's review of 884 generative-audio papers found that fewer than 10% discussed negative impacts, despite concerns such as fraud, deepfakes, and copyright infringement. This indicates an imbalance between capability research and impact research.

Second, there is no single accepted definition of content quality for generative media. Technical metrics can measure visual or audio properties but may not capture usefulness, cultural appropriateness, trust, or alignment with user intentions.

Third, many model studies evaluate systems under controlled conditions. Real production involves editing, human approval, copyright checks, and audience expectations. More research should therefore evaluate GenAI within realistic production workflows.

Fourth, further empirical research is needed on the long-term effects of GenAI on creative professionals. The technology may replace some tasks while creating new roles in prompting, editing, creative direction, and quality assurance.

Finally, ethical governance remains fragmented. Identifying ethical risks is not sufficient; practical mechanisms such as provenance, content labelling, consent systems, copyright management, and human review need to be evaluated in real-world settings.

Overall, the literature suggests that GenAI effectiveness should be assessed using a multidimensional framework combining technical quality, production efficiency, personalisation, user satisfaction, authenticity, ethical risk, and human oversight.

5. Conclusion

Generative AI has demonstrated significant potential to transform personalised video and audio production. The selected literature shows improvements in generation quality, multimodal capability, scalability, and production efficiency. At the same time, the studies reveal continuing limitations involving consistency, controllability, authenticity, copyright, privacy, bias, and creative labour.

The literature therefore supports a balanced view. GenAI should not be evaluated simply by asking whether it can generate content quickly or realistically. Effective use requires consideration of whether the content is relevant, trustworthy, ethically produced, and suitable for its intended audience. Human oversight remains important for creative direction, quality control, contextual judgement, and ethical responsibility.

Future research should develop evaluation frameworks that combine technical metrics with human assessment and ethical criteria. Research should also examine realistic production environments and the long-term impact of GenAI on creative work.

6. References

- [1] U. Singer et al., "Make-A-Video: Text-to-Video Generation without Text-Video Data," arXiv:2209.14792, 2022.
- [2] J. Ho et al., "Imagen Video: High Definition Video Generation with Diffusion Models," arXiv:2210.02303, 2022.
- [3] D. Kondratyuk et al., "VideoPoet: A Large Language Model for Zero-Shot Video Generation," in Proc. 41st Int. Conf. Machine Learning, vol. 235, pp. 25105–25124, 2024.
- [4] A. Agostinelli et al., "MusicLM: Generating Music From Text," arXiv:2301.11325, 2023.
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- [8] Y. Mirsky and W. Lee, "The Creation and Detection of Deepfakes: A Survey," ACM Computing Surveys, vol. 54, no. 1, 2021, doi: 10.1145/3425780.
- [9] T. Hagendorff, "Mapping the Ethics of Generative AI: A Comprehensive Scoping Review," Minds and Machines, vol. 34, 2024, doi: 10.1007/s11023-024-09694-w.
- [10] J. Amankwah-Amoah, S. Abdalla, E. Mogaji, A. Elbanna, and Y. K. Dwivedi, "The Impending Disruption of Creative Industries by Generative AI: Opportunities, Challenges, and Research Agenda," International Journal of Information Management, vol. 79, 102759, 2024, doi: 10.1016/j.ijinfomgt.2024.102759.

Note: The assessment permits reputable online sources in addition to academic literature. The final reference list should be expanded to approximately 20–30 sources, while the matrix can remain focused on the ten key studies selected for thematic synthesis.